Tony Di Fabbio, Dr.-Ing.
Date and Place of birth: 10.06.1995 Campobasso (CB), Italy
Residence: Via Exilles 84, 10146 Torino (TO), Italy
tonydifabbio@gmail.com @tonydifabbio Tony Di Fabbio

Experiences

05/2026 – ongoing
MSCA Global Postdoctoral Fellow (LINERFUN project) – Politecnico di Torino (beneficiary), University of Melbourne (outgoing host), ONERA (secondment)
Research on advanced acoustic liner technologies for Ultra-High Bypass Ratio engines, combining high-fidelity CFD/CAA, time-domain impedance boundary conditions, and physics-informed machine learning to design compact, low-drag, noise-reducing liners for aero-engines and future urban air mobility applications.

10/2024 – 04/2026
Postdoctoral Research Fellow at Leonardo Innovation Labs, Digital Twin and Advanced Simulation
Development of digital twin technologies for aerospace systems using high-fidelity CFD. Focus on aeroacoustics and high-speed compressible flows (supersonic/hypersonic), with applications to multiphysics simulations that support Multi-Disciplinary Design Optimization frameworks.

03/2024 – 06/2024
Scientific Researcher at The University of Melbourne
DAAD scholarship holder
Application of Machine Learning techniques—specifically Gene Expression Programming for Multi-Objective Optimization of complex engineering systems. Research focused on enhancing CFD-driven design processes through genetic algorithm modeling.

02/2020 – 02/2024
Scientific Researcher at Universität der Bundeswehr München
Numerical and theoretical research in aerodynamics and turbulence modeling. Activities included simulation and analysis of transonic cavity flows, leading-edge vortex dynamics, unsteady shock–vortex interactions, and control strategies for separated flows, with applications relevant to aerospace vehicle design and flow control.

10/2022 – 12/2022
Visiting Researcher at The University of Melbourne
Investigation of hybrid CFD–Machine Learning approaches for flowfield prediction and model acceleration. Focus on data-driven strategies for enhancing computational efficiency in fluid dynamics simulations.

04/2019 – 11/2019
Junior Scientific Researcher at DLR Lampoldshausen
Conducted high-fidelity simulations of a sub-scale LOX/CH_4 rocket combustor. Studied combustion stability and thermal characteristics to support experimental validation and propulsion system design.

02/2018 – 07/2018
Teaching Assistant at Politecnico di Torino
Assisted in lectures and practical sessions for the "Mechanical Machines" course. Supported students in understanding machine dynamics, thermodynamic cycles, and performance analysis.

10/2017 – 09/2018
Delegate and Student Representative at Collegio Einaudi, Torino
Acted as liaison between students and administrative board. Promoted academic initiatives and coordinated student engagement programs.

10/2017 – 07/2018
Student Team Member of Polito Sailing Team Torino
Contributed to designing and constructing a high-performance sailboat for the "MILLE E UNA VELA CUP 2017." Involved in hull optimization, structural layout, and system integration.

Education

02/2020 – 08/2024	**PhD** at the Universität der Bundeswehr München Final grade: Summa cum Laude Thesis title: *Efficient Turbulence Modelling for Vortical Flows from Swept Leading Edges*
09/2018 – 09/2019	**Technische Universität München** Master Student *Erasmus+ programme exchange*
09/2017 – 12/2019	**M.Sc. Politecnico di Torino** in Aerospace Engineering Final grade: 110 cum laude/110 Thesis title: *Numerical simulation of a sub-scale rocket combustor, carried out in the framework of CNES-DLR cooperation*
02/2017 – 06/2017	**École Centrale Paris** Bachelor Student *EU/Young Talents program with scholarship by Politecnico di Torino*
09/2014 – 07/2017	**B.Sc. Politecnico di Torino** in Aerospace Engineering Final grade: 110 cum laude/110 *Member of the Polito Young Talent Project: educational path of excellence for the top 200 undergraduate students in each academic year*
09/2015 – 07/2018	**Student at Collegio Universitario di Torino "R. Einaudi"** *The College "R. Einaudi" is legally recognized by the Italian Ministry of Education as "High cultural qualification Institution"*
09/2009 – 07/2014	**Liceo scientifico "A.Romita" Campobasso** *Second level college of science with final grade 100/100*

Skills

Languages	Mother tongue in Italian Strong reading, writing, and speaking competencies in English (C1) Good reading, writing, and speaking competencies in French, German (B1)
Coding	Python, C, C++, Matlab, LaTeX
Software	DLR-TAU code, OpenFoam, Ennova, Ansys Fluent, Comsol, Solidworks, Office
Field of Research	Delta-wing aerodynamics for high-agile military aircraft Applied CFD and turbulence modeling for aerodynamic applications Evolutionary Algorithms and Gene Expression Programming
Misc	Academic research, teaching, training, consultation Expertise in writing research proposals (MSCA 2024 Global Post-Doctoral Fellowship Seal of Excellence)
Social	Dynamic attitude inclined to approach problems and requests analytically and methodically to provide a solution soon. Strong orientation to teamwork and a sense of responsibility toward the aim. Optimistic and future-oriented.

Prizes, Awards and Funding Received

- POLITO Young Talent Project Member
- DGLR Young Talent Award (German Society for Aeronautics and Astronautics): €3,000
- EDISU University Scholarship: €25,000 (5 years, €5,000/year)
- Collegio Einaudi Merit Scholarships (3 awards): €1,000 each
- ERASMUS+ and POLITO Scholarship for TUM exchange: €6,000

- POLITO Young Talent Project Scholarship for École Centrale Paris exchange: €4,200
- Doctoral Fellowship from Airbus Defence and Space, Manching, Germany: €344,240.40
- DAAD Doctoral Fellowship for Research Abroad: €7,050
- LRZ SuperMUC-NG grants (2 awards): 60 million CPU hours (approx. €1,200,000 equivalent)
- MSCA Global Fellowship (awarded 2025; project start: May 2026), total funding: €309,153.72

Peer-Reviewed Journal Articles

- D. Kotlzowski, T. Di Fabbio, E. Tangermann, M. Klein, Y. Fang, and R. D. Sandberg, "Enhancing one-equation turbulence model for delta wing flows by gene expression programming," *AIAA Journal*, 2026.
  Contribution: Designed symbolic GEP corrections introducing nonlinear stress–strain terms and a revised turbulence production term; trained on a single delta wing and validated on double and triple delta configurations, enabling an interpretable and robust modelling workflow.
- T. Di Fabbio, E. Tangermann, M. Klein, Y. Fang, and R. D. Sandberg, "Strategies for Enhancing One-Equation Turbulence Model Predictions Using Gene-Expression Programming," *Fluids*, 2024, 9(8).
  Contribution: Developed two GEP pathways extending the Boussinesq assumption and constructing a new one-equation closure using solver-in-the-loop training for deployable wall-bounded turbulence models.
- K. Rajkumar, T. Di Fabbio, E. Tangermann, and M. Klein, "Physical aspects of vortex–shock dynamics in delta wing configurations," *Physics of Fluids*, 2024.
  Contribution: Quantified shock–vortex interaction mechanisms across double and triple delta planforms through CFD and enstrophy transport analysis, identifying geometry- and incidence-driven buffet sensitivity.
- T. Di Fabbio, K. Rajkumar, E. Tangermann, and M. Klein, "Towards the understanding of vortex breakdown for improved RANS turbulence modelling," *Aerospace Science and Technology*, 2024.
  Contribution: Diagnosed physical and modelling causes of one-equation RANS failures in transonic sideslip and proposed a validated modification improving stability and accuracy.
- T. Di Fabbio, E. Tangermann, and M. Klein, "Analysis of the vortex-dominated flow field over a delta wing at transonic speed," *The Aeronautical Journal*, 2023.
  Contribution: Produced high-fidelity datasets resolving vortex systems and shocks; analysed eddy viscosity and Reynolds stresses in vortex formation and maintenance.
- T. Di Fabbio, E. Tangermann, and M. Klein, "Investigation of transonic aerodynamics on a triple-delta wing in sideslip conditions," *CEAS Aeronautical Journal*, 2022.
  Contribution: Demonstrated RANS limitations and advantages of high-fidelity approaches for predicting vortex and shock interactions, improving aerodynamic load predictions.

Conference Presentations and Proceedings

- G. A. Bres, C. B. Ivey, L. Heidt, K. Wang, S. T. Bose, R. Hottois, J. Guerrero, and T. Di Fabbio, "GPU-Accelerated Large-Eddy Simulations of High-Speed Cavity Flows With Dynamic Doors Deployment," 32nd AIAA/CEAS Aeroacoustics Conference, 2026, pp. 3359.
- E. Segalerba, T. Di Fabbio, V. Rossano, and J. Guerrero, "Towards a generalized open-source turbulence model for external aerodynamics," 20th OpenFOAM Workshop (OFW20), Vienna, Austria, 2025.
- V. Rossano, T. Di Fabbio, E. Segalerba, and J. Guerrero, "Verification and Validation of OpenFOAM for High-Speed Aerodynamics," 20th OpenFOAM Workshop (OFW20), Vienna, Austria, 2025.

- G. Zampa, T. Di Fabbio, E. Segalerba, and J. Guerrero, “Optimization of Hypersonic Re-Entry Vehicle Aerodynamics for Communication Blackout Mitigation,” European Microwave Week, Utrecht, The Netherlands, 2025.
- T. Di Fabbio, E. Segalerba, E. Tangermann, and M. Klein, “Towards a machine learning-augmented one-equation turbulence model for external aerodynamics in an open-source framework,” DGLR Conference, Augsburg, Germany, 2025 (forthcoming).
- T. Di Fabbio, E. Tangermann, and M. Klein, “Flow pattern analysis on a delta wing at transonic speed,” AIAA SciTech Forum, San Diego, CA, 2022.
- T. Di Fabbio, E. Tangermann, and M. Klein, “Investigation of leading-edge vortices on a triple-delta-wing configuration using scale-adaptive simulation,” ICAS Congress, Stockholm, Sweden, 2022.
- T. Di Fabbio, E. Tangermann, and M. Klein, “Investigation of Reynolds stresses prior to vortex breakdown on a triple-delta wing at transonic condition,” 12th International Symposium on Turbulence and Shear Flow Phenomena (TSFP12), Osaka, Japan, 2022.
- T. Di Fabbio, E. Tangermann, M. Klein, S. Ketterl, and A. Winkler, “Comparative scale-resolving and RANS simulations of a delta wing configuration,” DGLR Conference, Bonn, Germany, 2020.